**4** **(a)** Define specific heat capacity.

.....

.....

..... [2]

**(b)** A volume of 0.0024 $m^3$ of oil is heated in a pan for 7.0 min. The temperature of the oil increases from 20 °C to 180 °C.

The density of the oil is 910 kg / $m^3$. The specific heat capacity of the oil is 2000 J / (kg °C).

**(i)** Calculate the mass of oil in the pan.

mass = ..... [2]

**(ii)** Calculate the energy required to increase the temperature of the oil.

energy = ..... [2]

**(iii)** Calculate the power required to supply the energy calculated in **(b)(ii)**.

power = ..... [2]

[Total: 8]

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